DET-008 | Suspicious Scheduled Task Creation

ID: DET-008
MITRE ID: T1053.005
Data Source: DeviceProcessEvents

Severity: High
Tactic:Persistence
Category: Endpoint

ADX MAPPING LOGIC:
Mapping: Source = 'schtasks.exe process', EventType = 'task action binary', State = 'DeviceName'. We use countif() and sumif() to flag high-frequency task creations — practicing aggregate conditional functions.

OBJECTIVE: Practice countif() and sumif() — conditional aggregation operators. Real Sentinel: schtasks.exe /create with powershell/cmd/wscript in the action field pointing to suspicious paths like AppData or Temp.

KQL QUERY (RUNNABLE ON ADX FREE TIER)

```
StormEvents
| where StartTime > ago(365d)
| summarize
TotalTasks = count(),
SuspiciousTasks = countif(EventType in (
"Tornado","Waterspout","Dust Devil" // = suspicious binaries
)),
HighDamageTasks = countif(DamageProperty > 50000),
TotalDamage = sumif(DamageProperty, DamageProperty > 0),
PeakDamage = max(DamageProperty),
TaskTypes = make_set(EventType, 5),
FirstTask = min(StartTime),
LastTask = max(StartTime)
by State, Source
| extend
SuspiciousRatio = round(100.0 * SuspiciousTasks / TotalTasks, 1),
RiskLevel = iff(SuspiciousTasks > 5, "HIGH",
iff(SuspiciousTasks > 2, "MEDIUM", "LOW"))
| where SuspiciousTasks > 0
| project State, Source, TotalTasks, SuspiciousTasks, SuspiciousRatio,
RiskLevel, TotalDamage, PeakDamage,TaskTypes, FirstTask, LastTask
| order by SuspiciousTasks desc | take 20
```

INVESTIGATION STEPS
1. Review task action — what binary/script does it execute?
2. Check the scheduled time and frequency.
3. Identify the user context in which the task runs.
4. Verify if the task was created by a legitimate admin.

RESPONSE ACTIONS

- Delete the suspicious scheduled task immediately.
- Investigate what the task action would have executed.
- Check if the same task exists on other endpoints.
- Review the account that created the task for compromise.

Output from Above KQL Query:

```
StormEvents
| where StartTime > ago(365d)
| summarize TotalTasks = count(),
SuspiciousTasks = countif(EventType in ("Tornado","Waterspout","Dust Devil" )),
HighDamageTasks = countif(DamageProperty > 50000),
TotalDamage = sumif(DamageProperty, DamageProperty > 0),
PeakDamage = max(DamageProperty),
TaskTypes = make_set(EventType, 5),
FirstTask = min(StartTime),
LastTask = max(StartTime) by State, Source
| extend
SuspiciousRatio = round(100.0 * SuspiciousTasks / TotalTasks, 1),
RiskLevel = iff(SuspiciousTasks > 5, "HIGH",
iff(SuspiciousTasks > 2, "MEDIUM", "LOW"))
| where SuspiciousTasks > 0
| project State, Source, TotalTasks, SuspiciousTasks, SuspiciousRatio,
RiskLevel, TotalDamage, PeakDamage, TaskTypes, FirstTask, LastTask
| order by SuspiciousTasks desc
| take 20
```

Table 1 | Add visual | Stats | 2026-0

StartTime	DeviceName	ProcessProxy	EpisodeIdStr	SimBase64	B64Length	ObfuscationRisk

No Rows To Show

KQL Detection Query in Sentinel:

```
DeviceProcessEvents
| where TimeGenerated > ago(1h)
| where FileName in~ ("schtasks.exe","at.exe")
| where ProcessCommandLine has "/create"
| extend TaskAction = extract(
@"/tr\s+[""]?([^\s""]+)", 1, ProcessCommandLine)
| extend TaskName = extract(
@"/tn\s+[""]?([^\s""]+)", 1, ProcessCommandLine)
| where TaskAction has_any (
"powershell","cmd","wscript","cscript",
"mshta","rundll32","temp","appdata","\\Users\\")
| project TimeGenerated, DeviceName, AccountName,
ProcessCommandLine, TaskName, TaskAction,
InitiatingProcessFileName
| order by TimeGenerated desc
```